0. $E' \to E\$$
1. $E \to E+T$
2. $E \to T$
3. $T \to (E)$
4. $T \to a$

FIRST

$E' \to \{ (, a \}$
$E \to \{ (, a \}$
$T \to \{ (, a \}$

$\underline{I_0}$ = closure ([$E' \to .E$, \$])
= { [$E' \to .E$, \$],
[$E \to .E+T$, FIRST(E\$) = {\$}]
[$E \to .T$, \$],
[$T \to .(E)$, \$],
[$T \to .a$, \$],
[$E \to .E+T$, FIRST(+T\$) = {+}],
[$E \to .T$, +],
[$T \to .(E)$, +],
[$T \to .a$, +] }
= {[$E' \to .E$, \$],
[$E \to .E+T$, \$/+],
[$E \to .T$, \$/+],
[$T \to .(E)$, \$/+],
[$T \to .a$, \$/+] }

$\underline{I_1}$ = goto (I_0 , E)
= { [$E' \to E.$, \$],
[$E \to E.+T$, \$/+] }

$\underline{I_2}$ = goto (I_0 , T)
= { [$E \to T.$, \$/+] }

$\underline{I_3} = goto(I_0, ()$

$= \{[T \to (.E), \$],$

$[T \to (.E), +],$

$[E \to .E+T,)],$ FIRST($)\$) = {)}, FIRST(+)) = {)}

$[E \to .T,)].$

$[T \to .(E),)],$

$[T \to .a,)],$

$[E \to .E+T, FIRST(+T)) = \{+\}],$

$[E \to .T, +],$

$[T \to .(E), +],$

$[T \to .a, +]\}$

$= \{[T \to (.E), \$/+],$

$[E \to .E+T, +/)],$

$[E \to .T, +/)],$

$[T \to .(E), +/)],$

$[T \to .a, +/)]\}$

$\underline{I_4} = goto(I_0, a)$

$= \{[T \to a., \$/+]\}$

$\underline{I_5} = goto(I_1, +)$

$= \{[E \to E+.T, \$/+],$

$[T \to .(E), \$/+],$

$[T \to .a, \$/+]\}$

$\underline{I_6}$ = goto (I_3, E)

$= \{ [T \to (E \cdot), \$/+],$

$[E \to E \cdot + T, +/)] \}$

$\underline{I_7}$ = goto (I_3, T)

$= \{ [E \to T \cdot, +/)] \}$

$\underline{I_8}$ = goto $(I_3, ($ $)$

$= \{ [T \to (\cdot E),)],$

$[T \to (\cdot E), +],$

$[E \to \cdot E+T, \text{FIRST}(\,))) = \{)\}],$

$[E \to \cdot T,)],$

$[T \to \cdot (E),)],$

$[T \to \cdot a,)],$

$[E \to \cdot E+T, \text{FIRST}(+)) = \{+\}],$

$[E \to \cdot T, +]$

$[T \to \cdot (E), +]$

$[T \to \cdot a, +] \}$

$= \{ [T \to (\cdot E),)/+],$

$[E \to \cdot E+T,)/+],$

$[E \to \cdot T,)/+],$

$[T \to \cdot (E),)/+],$

$[T \to \cdot a,)/+] \}$

$\underline{I_9}$ = goto (I_3, a)

$= \{ [T \to a \cdot, +/)] \}$

$\underline{I_{10}}$ = goto (I_5, T)

$= \{ [E \to E+T \cdot, \$/+] \}$

$\rangle$ goto (I_5, ( )

$= \boxed{I_3}$

$\rangle$ goto (I_5 , a)

$= \boxed{I_4}$

$\underline{I_{11}}$ = goto (I_6,))

$= \{ [T \rightarrow (E) \bullet, \$ / +] \}$

$\underline{I_{12}}$ = goto (I_6, +)

$= \{ [E \rightarrow E + \bullet T, + /)],$

$[T \rightarrow \bullet (E), + /)],$

$[T \rightarrow \bullet a , + /)] \}$

$\underline{I_{13}}$ = goto (I_8, E)

$= \{ [T \rightarrow (E \bullet), + /)],$

$[E \rightarrow E \bullet + T , + /)] \}$

$\rangle$ goto (I_8, T)

$= \boxed{I_7}$

$\rangle$ goto (I_8, ( )

$= \boxed{I_8}$

$\rangle$ goto (I_8, a)

$= \boxed{I_9}$

$\underline{I_{14}}$ = goto (I_{12}, T)

$= \{ [E \rightarrow E + T \bullet , + /)] \}$

> goto $(I_{12}, ()$

$= \boxed{I_8}$

> goto (I_{12}, a)

$= \boxed{I_9}$

$\underline{I_{15}} = \text{goto}(I_{13},))$

$= \{ [T \to (E)\bullet, + /)] \}$

> goto $(I_{13}, +)$

$= \boxed{I_{12}}$

Ahora, se compactan los estados para obtener el autómata LALR(1):

$I_0 \to$ — No se compacta

$I_1 \to$ —

$I_2 \to I_2 \cup I_7 = \{ [E \to T\bullet, \$ / + /)] \}$

$I_3 \to I_3 \cup I_8 = \{ [T \to (\bullet E), \$ / + /)]$
$[E \to \bullet E+T, + /)]$
$[E \to \bullet T, + /)]$
$[T \to \bullet (E), + /)]$
$[T \to \bullet a, + /)] \}$

$I_4 \to I_4 \cup I_9 = \{ [T \to a\bullet, \$ / + /)] \}$

$I_5 \to I_5 \cup I_{12} = \{ [E \to E+\bullet T, \$ / + /)],$
$[T \to \bullet (E), \$ / + /)],$
$[T \to \bullet a, \$ / + /)] \}$

$I_6 \rightarrow I_6 \cup I_{13} = \{ [T \rightarrow (E \bullet), \$/+/)], [E \rightarrow E \bullet + T, +/)] \}$

$I_7 \rightarrow X$ Desaparece, se compacto con otro

$I_8 \rightarrow X$

$I_9 \rightarrow X$

$I_{10} \rightarrow I_{10} \cup I_{14} = \{ [E \rightarrow E + T \bullet, +/)/\$] \}$

$I_{11} \rightarrow I_{11} \cup I_{15} = \{ [T \rightarrow (E) \bullet, \$/+/)] \}$

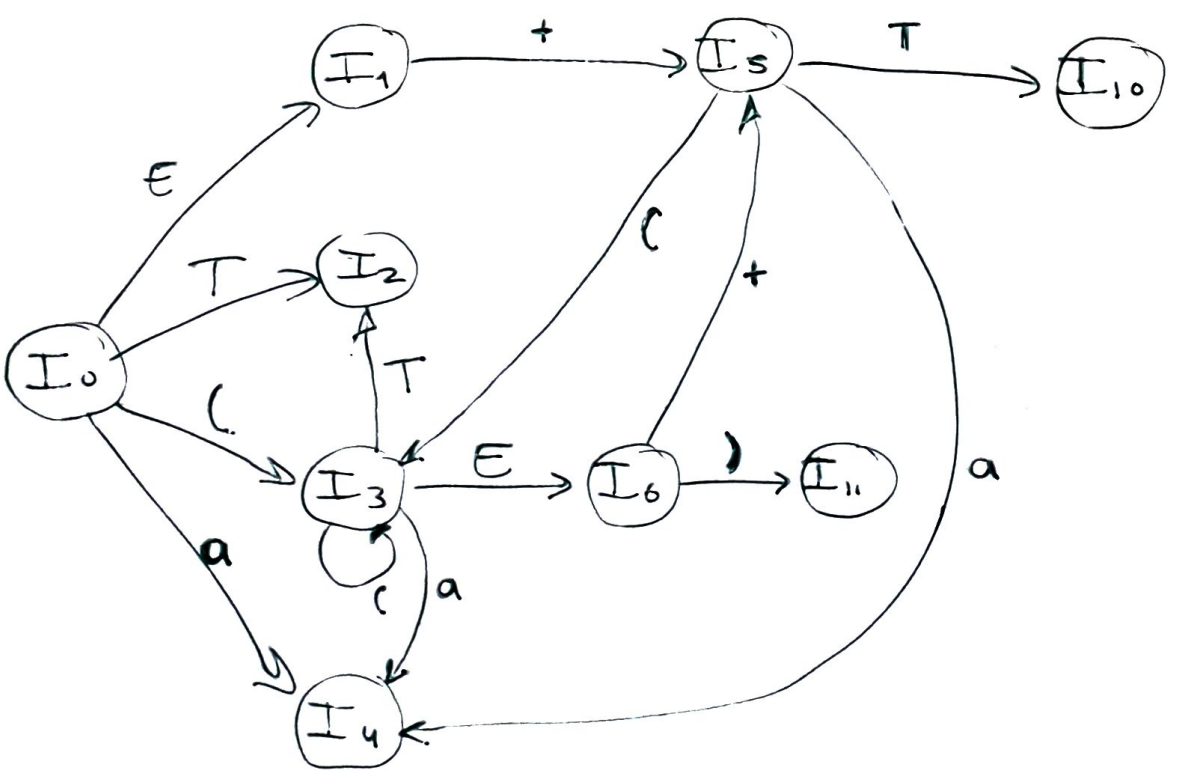
$I_{12} \rightarrow X$

$I_{13} \rightarrow X$

$I_{14} \rightarrow X$

$I_{15} \rightarrow X$

Autómata:



Cadenas aceptadas	Cadenas no-aceptadas
(a+a)	a+
a+a+a	((a
(a)	a a